

Homework 02: Chapter 03

EEC 3150

January 24, 2023

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (10 points, Python)

Write a program which will convert a binary integer (i.e., 1001011) taken from the input() function into its decimal equivalent by adding multiples of powers of 2.

Exercise 2 (10 points, Python)

Write a program that will use brute force (i.e., exhaustive enumeration) to find an approximation of $\log_{10}(42)$, i.e., that will solve $f(x) = 10^x - 42 = 0$. Set your ϵ and step as you best see fit.

Exercise 3 (10 points, Python)

Write a program that will use bisection to find an approximation of $\log_{10}(42)$, i.e., that will solve $f(x) = 10^x - 42 = 0$. Set your ϵ , high, and low as you best see fit.

Exercise 4 (10 points, Python)

Write a program that will use Newton-Raphson to find an approximation of $\log_{10}(42)$, i.e., that will solve $f(x) = 10^x - 42 = 0$. Set your ϵ and initial guess as you best see fit. Also, since you will need to evaluate $\ln(10)$, use a variable `log10 = 2.3025` to approximate it. (We can easily evaluate logarithms with Python, but we need to import additional libraries and we haven't discussed that yet.)

Exercise 5 (*10 points, Short Answer*)

Please describe the relative pros and cons of the following methods:

1. Brute force
2. Bisection
3. Newton-Raphson